

Revisiting the Mechanical Advantage of Compliant Mechanisms

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Abstract. In this work, the foundational work on the mechanical advantage of compliant mechanisms by Salamon and Midha (1998) is revisited. Inspired by this work, some general relations are derived using kinetoelastic equations. Furthermore, the interpretation of types of mechanical advantages is clarified to avoid ambiguities and to address some shortcomings. The mechanical advantage for a general deformation of a single-input single-output compliant mechanism is derived using the tangent stiffness/compliance matrix. It is simplified for linearised deformations, and the relationships are expressed in terms of the eigenvalues and eigenvectors of the stiffness matrix. These results and their interpretation are elucidated with the examples of a compliant crimping mechanism and a compliant four-bar linkage.

Keywords: Compliant mechanisms, Rigid-body mechanisms, Displacement advantage, Mechanical advantage

1 Introduction

Compliant mechanisms are a relatively recent addition to the arsenal of engineers. These mechanisms derive their degrees of freedom through the elastic deformation of members as opposed to only kinematic joints as in a rigid-body mechanism. The coupled contribution of both kinematics and the elastic deformation characteristics of the mechanism (together termed *kinetoelastics*) influences the output characteristics of the mechanism. The two most popular measures used to quantify the performance of a mechanism are its displacement advantage (DA) and mechanical advantage (MA). The former is the ratio of displacements at the designated output and input, and the latter is the ratio of forces at the designated output and input.

It is important to note the subtleties in the definition when applied to rigid body mechanisms and compliant mechanisms. For rigid body mechanisms, these measures are purely dependent on the instantaneous geometric configuration of the mechanisms. and can be quantified by computing the instant centre(s). However, as will be explained later, such computation is not possible for a compliant mechanism in general. From [1], it can be understood that the MA of a single-input single-output compliant mechanism at an instant cannot be stated uniquely, and it can vary based on the forces and kinematic constraints imposed on the system. For example, MA_1 , which refers to MA computed for a fixed output displacement, is a function of the input force. Thus, it cannot be

stated uniquely for a compliant mechanism in the same way as MA can be calculated for a rigid-body mechanism. To define MA and DA uniquely (for small deformations) for a given compliant mechanism, we must specify at least two of the following four parameters: $\{f_i, f_o, u_i, u_o\}$ which refer to input force, output force, input displacement, and output displacement, respectively. This is also true for a rigid-body mechanism, but we implicitly consider the case of u_i and u_o being zero for the definition of mechanical advantage (a consequence of static equilibrium) and the case f_i and f_o being zero for the definition of displacement advantage (mobility and degree of freedom) in a rigid-body mechanism.

The ability of compliant mechanisms to transmit forces is pivotal in several active areas of research, including soft robotics, MEMS, biomedical devices and precision instrumentation. Its contemporary relevance warrants a detailed study of the mechanical advantage of a compliant mechanism. To this end, a general analysis of the mechanical advantage of a compliant mechanism based on kinetoelastics is attempted in this work.

1.1 Mechanical advantage of compliant mechanisms based on the energy method

In their seminal work on the mechanical advantage of compliant mechanisms, Salamon and Midha (1998) derived relations for the mechanical advantage based on energy principles. They had ended up with the following relation:

$$\text{MA} = \frac{f_o}{f_i} = \frac{\Delta u_i}{\Delta u_o} - \frac{\Delta U}{f_i \Delta u_o} \equiv \text{MA}_r - \text{MA}_c, \quad (1)$$

where f_o and f_i are the respective forces at the output (o) and input (i) (along the designated direction); Δu_o and Δu_i are the incremental displacements applied at the respective ports, and ΔU is the change in strain energy owing to the incremental displacement. In their work, the first term was attributed as the mechanical advantage of an equivalent “rigid-body” (as the term appears similar to the instantaneous velocity-based relation obtained for rigid-body mechanisms). They had also remarked that a single equivalent “rigid-body” cannot exist for the entire range of motion of a compliant mechanism and that the dimensions of this “rigid body” vary as the mechanism deforms. They then attributed the second term to the loss in mechanical energy due to compliance.

Such an interpretation, though very meaningful, has the following shortcomings. In general, there is no known way to perform instant centre analysis for a general compliant mechanism. Thus, it is not possible to calculate MA_r just from the geometry of the mechanisms, nor can we construct an equivalent “rigid-body” against which the compliant mechanisms can be compared. Also, it is possible for MA_c to be negative without violating the intended purpose of a compliant mechanism [2].

2 Mechanical advantage of compliant mechanisms based on kinetoelastics

The relation for MA would have more utility if it conveyed information in terms of its instantaneous stiffness matrix rather than its incremental strain energy, as the former is more intuitive to estimate/understand in practice. To achieve this, a new relation for MA based on the kinetoelastics is presented here.

In general, a single stiffness matrix cannot accurately represent the deformation characteristics for the entire range of motion. Thus, a tangent stiffness-based approach is adopted here to account for both geometric and material nonlinearities. The tangent stiffness matrix ($[\mathbf{K}_t(\mathbf{u})]$)/tangent compliance matrix ($[\mathbf{C}_t(\mathbf{u})]$) depends on the deformed configuration of the mechanism at an instant. It relates the incremental force ($\{\Delta \mathbf{f}\}$) applied to the ensuing incremental displacement ($\{\Delta \mathbf{u}\}$) as stated below:

$$[\mathbf{K}_t(\mathbf{u})]\{\Delta \mathbf{u}\} = \{\Delta \mathbf{f}\} \iff \{\Delta \mathbf{u}\} = [\mathbf{C}_t(\mathbf{u})]\{\Delta \mathbf{f}\} \quad (2)$$

Here, the tangent stiffness matrix is assumed invertible in the entire range of motion. Then the incremental displacement of the input and output can be related to the incremental input and output force (considered in the direction opposite to the desired output displacement) as stated below:

$$\Delta u_i = [C_t(\mathbf{u})]_{ii} \Delta f_i - [C_t(\mathbf{u})]_{io} \Delta f_o \quad (3)$$

$$\Delta u_o = [C_t(\mathbf{u})]_{oi} \Delta f_i - [C_t(\mathbf{u})]_{oo} \Delta f_o \quad (4)$$

We can rearrange Eq. (4) as

$$\Delta f_o = \frac{[C_t(\mathbf{u})]_{oi} \Delta f_i}{[C_t(\mathbf{u})]_{oo}} - \frac{\Delta u_o}{[C_t(\mathbf{u})]_{oo}} \quad (4a)$$

By integrating Eq. (4a) from the initial stress-free state ($1 \equiv \{u_i = 0, u_o = 0, f_i = 0, f_o = 0\}$) to its final state ($2 \equiv \{u_i = u_i^*, u_o = u_o^*, f_i = f_i^*, f_o = f_o^*\}$) along a prescribed path, we obtain

$$f_o^* = \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} - \int_1^2 \frac{du_o}{[C_t(\mathbf{u})]_{oo}} \quad (5)$$

The preceding expression can be rewritten in the following form (by dividing the previous equation by $f_i^* = \int_1^2 df_i$):

$$\frac{f_o^*}{f_i^*} = \text{MA}_s \left(1 - \frac{f_n}{f_i^*} \right) \quad (6)$$

$$f_n \equiv \left(\frac{\int_1^2 \frac{du_o}{[C_t(\mathbf{u})]_{oo}}}{\frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}}} \right), \quad \text{MA}_s = \frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}}$$

Now, let us consider a compliant mechanism in its stress-free state, which is loaded by a quasi-statically applied input force while the mechanism covers some

output displacement (u_o^*) before the workpiece (assumed rigid) can be engaged. Let us define the intermediate stage where the mechanism begins to engage the workpiece as state $m \equiv \{u_i^m, u_o^*, f_i^m, f_o = 0\}$. Then Eq. (6) becomes

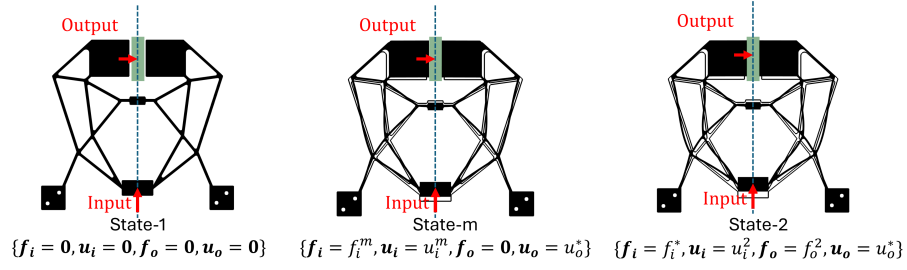
$$\begin{aligned} \frac{f_o^*}{f_i^*} &= \frac{1}{f_i^*} \int_1^2 \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} - \frac{1}{f_i^*} \int_1^m \frac{[C_t(\mathbf{u})]_{oi} df_i}{[C_t(\mathbf{u})]_{oo}} \\ &= \frac{1}{f_i^*} \int_{f_i^m}^{f_i^*} \frac{[C_t(\mathbf{u})]_{oi} dF_i}{[C_t(\mathbf{u})]_{oo}}, \end{aligned} \quad (7)$$

where f_i^m is the input force at which the mechanism engages the workpiece. The same is pictorially described in Fig. 1a.

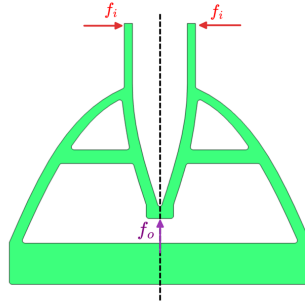
If the output displacement ($u_o^* = 0$) were to be constrained during the entire motion, then the force loss due to the second term (f_n) is absent, and essentially the mechanical advantage is maximised if the mechanism behaves like a structure (i.e. the corresponding degree of freedom [3] is arrested). We note that the first term MA_s is the path average of the instantaneous MA_r during the motion. Also, in this definition, f_n is strictly non-negative. To further highlight this, the variability of f_n , MA_s , MA_r , and f_c for a symmetric half-section of a compliant crimping mechanism (shown in Fig. 1b) at high loads is shown in figs. 1c and 1d. In the former, u_o^* is zero, i.e. the mechanism is engaged with the workpiece in its stress-free state and in the latter, u_o^* is set to 2 mm. In both cases, we note that at high loads, compliance can be used to contribute to MA.

And based on Eq. (6), we can also understand how $MA_1(f_i^*)|_{u_o^*}$ and $MA_2(u_o^*)|_{f_i^*}$ (defined as the variation of mechanical advantage with output displacement for a fixed input force) behaves for very small deformation and very large deformation. For small deformation, MA_1 varies with the input force as a rectangular hyperbola, and for the case u_o^* is zero, the MA_1 is constant [1, 4]. This, however, is not true at large loads and the mechanism can converge to a smaller or even zero mechanical advantage based on the way the ratio $\frac{[C_t(\mathbf{u})]_{oi}}{[C_t(\mathbf{u})]_{oo}}$ scales with load. As for MA_2 , it will decrease linearly at small deformations [1, 4], but can decay superlinearly/sublinearly at very high deformation depending on how the denominator term ($[C_t(\mathbf{u})]_{oo}$) in the second integral term scales with deformation along the prescribed path. These details can be noticed in figs. 1e and 1f (linear elastic material properties are assumed for calculation, $E = 3$ GPa and $\nu = 0.35$).

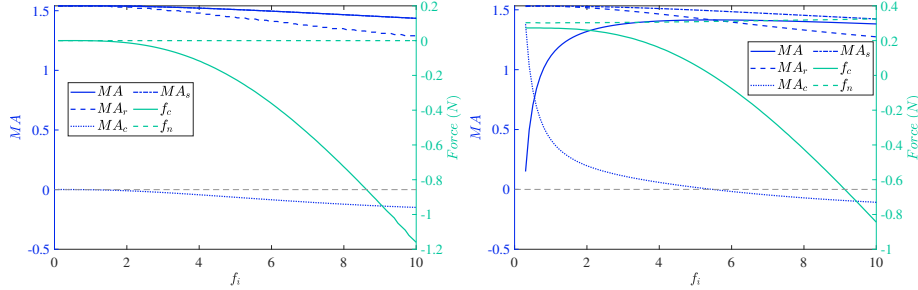
It can be summarised that the mechanical advantage in general is maximum when the mechanism behaves like a structure (output port is fixed), and there is a loss in mechanical advantage due to the deformation the system must experience to engage the workpiece. The mechanical advantage of the structure at the particular load is the average of the ratio of incremental displacement of input and output (MA_r) (output displacement incremented at constant input force) over the entire motion.



(a) The force-displacement path considered in analysis here, the mechanism engages the workpiece at an input force of f_i^m where the output displacement is u_o^*



(b) A compliant crimping mechanism, CAD model is available in our compliant mechanisms database (<https://mecheng.iisc.ac.in/m2d2/CMcollection/>).



(c) Variability of the different mechanical advantage terms and compliance forces with input force for $u_o^* = 0$

(d) Variability of the different mechanical advantage terms and compliance forces with input force for $u_o^* = 2$ mm

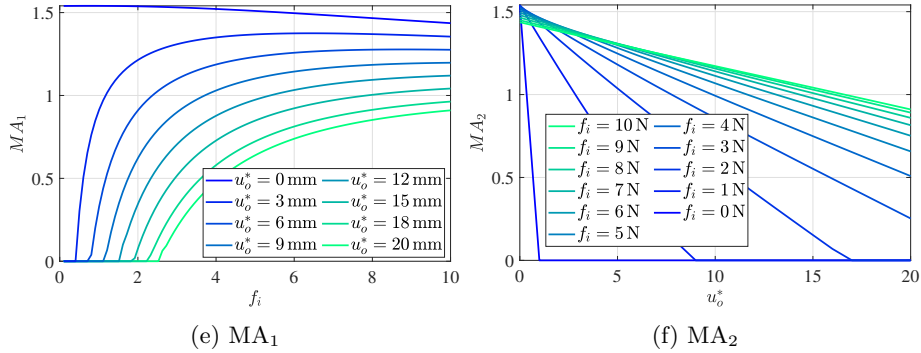


Fig. 1: On the mechanical advantage of compliant mechanisms

3 Mechanical advantage for a small deformation

For small deformation, the deformation characteristics of the continuum structure can be represented by the initial stiffness matrix.

$$[\mathbf{K}]\{\mathbf{u}\} = \{\mathbf{f}\} \quad (8)$$

The matrix $[\mathbf{K}]_{n \times n}$ is symmetric and positive semi-definite in general, thus eigenvalues ($\lambda_k \forall k \in \{1, 2..n\}$) can be computed along with orthonormal eigenvectors ($\mathbf{a}_k \forall k \in \{1, 2..n\}$). Thus $\{\mathbf{u}\}$ can be calculated as:

$$\{\mathbf{u}\} = \sum_{k=1}^n \frac{(\mathbf{a}_k \otimes \mathbf{a}_k)\{\mathbf{f}\}}{\lambda_k} \quad (9)$$

Let us consider the application of only two forces¹ in the body, one about the input and the other about the output, then the input and output displacements are as follows:

$$u_i = \sum_{k=1}^n \frac{f_i(a_k \otimes a_k)_{ii}}{\lambda_k} - \frac{f_o(a_k \otimes a_k)_{io}}{\lambda_k} \quad (10)$$

$$u_o = \sum_{k=1}^n \frac{f_i(a_k \otimes a_k)_{io}}{\lambda_k} - \frac{f_o(a_k \otimes a_k)_{oo}}{\lambda_k} \quad (11)$$

The last relation can be simplified to:

$$\frac{f_o}{f_i} = \text{MA}_s \left(1 - \frac{f_n}{f_i} \right) \quad (12)$$

$$f_n \equiv \frac{u_o}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}, \quad \text{MA}_s = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{oo}}{\lambda_k}} \quad (13)$$

Once again, it can be noticed that while linearising the deformation characteristics in the stress-free state, the first term corresponds to the mechanical advantage of the structure (referred to as MA_s) obtained by fixing the output displacement ($u_o = 0$). And there is a loss in MA when the output port must cover a finite distance before engaging the workpiece.

Similarly, it is possible to represent the displacement advantage as:

$$\frac{u_o}{u_i} = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{ii}}{\lambda_k}} \left(1 - \frac{u_c}{u_i} \right) \quad (14)$$

$$u_c \equiv \frac{f_o \left(\left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{ii}}{\lambda_k} \right) \left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{oo}}{\lambda_k} \right) - \left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k} \right)^2 \right)}{\left(\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k} \right)} \quad (15)$$

¹ The output force is considered in the direction opposite to that of the output displacement to be consistent with the intended purpose of a mechanism.

Here, the first term is that of the mechanism in the absence of output force, and the second term is due to the loss in displacement advantage due to the action of the output force present, if any kinematic constraint on the output displacement is prescribed. Similar to MA_s , we will define DA_f as the displacement advantage in the absence of any constraints on the output displacement.

DA_f can be rewritten as:

$$DA_f = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{ii}}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_i}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k}{\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k}}, \quad (16)$$

where DA_k is the ratio $(a_k)_o/(a_k)_i$ for k^{th} eigenvector. It is to be noted that the terms in the denominator are always non-negative and that the DA of the actual mechanism is a weighted average of the displacement advantage of the individual eigenvectors. Similarly, we can rewrite MA_s as:

$$MA_s = \frac{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{io}}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k \otimes a_k)_{oo}}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_i \times (a_k)_o}{\lambda_k}}{\sum_{k=1}^n \frac{(a_k)_o \times (a_k)_o}{\lambda_k}} = \frac{\sum_{k=1}^n \frac{(a_k)_o^2}{\lambda_k} \left(\frac{1}{DA_k} \right)}{\sum_{k=1}^n \frac{(a_k)_o^2}{\lambda_k}} \quad (17)$$

We once again notice that the mechanical advantage, in this case, can be represented as the weighted average of the inverse of the displacement advantage of each eigenvector (but the weights are different here). It can also be represented as follows:

$$MA_s = \frac{\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k}{\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k^2}, \quad (18)$$

and consequently:

$$MA_s \times DA_f = \frac{\left(\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k \right)^2}{\left(\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} DA_k^2 \right) \left(\sum_{k=1}^n \frac{(a_k)_i^2}{\lambda_k} \right)} \leq 1 \quad (19)$$

The equality in the preceding inequality is only satisfied if all the mode shapes have the same displacement advantage (which is not allowed, as the eigenvectors must form a basis) or if one of the contributing eigenvalues tends to zero (indicating rigid body mechanism²).

Thus, in general:

$$DA \times MA \leq DA_f \times MA_s < 1 \quad (20)$$

Also, from eqs. (16) and (17), if the designated input and output are flipped, then the MA_s and DA_f are merely swapped (similar to rigid body mechanisms). Thus, the rest of the analysis will only focus on the displacement advantage for linear studies, as it is more intuitive and easier to visualise in comparison to the mechanical advantage in the eigenvector framework.

² Here, the definition of MA_s and DA_f are analogous to how MA and DA are defined for a rigid mechanism.

3.1 Compliant 4-bar linkage

It is instructive to consider the case of a compliant 4-bar linkage to appreciate the representation in the eigenvector framework. There are several synthesis methods which rely on the use of the compliant 4-bar as a building block. One of the popular methods to do this is the instant centre-based synthesis method [5]. Where the mechanism is idealised as a rigid-body equivalent, and synthesis is performed based on the estimates of the displacement and mechanical advantage of this rigid-body equivalent. Thus, it is of interest to understand the displacement advantage of this mechanism in comparison to its equivalent rigid body linkage (obtained from replacing the beam joint with pin joints as done in [5]).

Here, the illustrative example of two compliant 4-bar linkages with identical displacement advantage in their rigid-body linkage representation, shown in Fig. 2a, is considered. Their displacement advantage (calculated with Euler-Bernoulli beam theory-based finite element formulation) is plotted against the slenderness ratio in Fig. 2b. It can be noted that the values of both mechanisms are different and smaller than the rigid-body equivalent (DA_{IC}). Information about this can be inferred from their eigenvectors and eigenvalues denoted in figs. 2c to 2f. From these figures, it is seen that only three out of the six eigenvectors contribute meaningfully towards the actual displacement advantage. It is instructive to note that some eigenvectors (most significantly green) root against the deformation desired from the instant centre calculation. This leads to both mechanisms having their displacement advantage less than the equivalent rigid-body linkage. We also note that the most compliant as well as the third most compliant eigenvector roughly corresponds to that of the instant centre equivalent, and that the difference in performance between mechanisms can be attributed to the most compliant eigenvector.

3.2 Combining compliant mechanisms

The building block-based synthesis methods involve the composition of several independently designed mechanisms. In the case of rigid body mechanisms, the performance characteristics DA and MA of a composition of rigid body mechanisms are merely the product of the individual DA and MA of each mechanism. However, the same is not true for compliant mechanisms. This can be understood by representing two compliant mechanisms in terms of their spring lever parameters k_i^1 , k_o^1 , n_1 and k_i^2 , k_o^2 , n_2 [6]. Where k_i denotes the input stiffness when output is unconstrained and k_o denotes the output stiffness in the absence of input force, and n represents the amplification factor (ratio of output displacement and input displacement when the former is unconstrained). If the input of mechanism-2 is connected (arresting only the degree of freedom of interest) to the output of mechanism-1. Then, the equivalent DA_f for this mechanism is:

$$n^* = \left(\frac{k_o^1}{k_i^2 + k_o^1} \right) n_1 n_2 \quad (21)$$

Thus, the DA (or MA) obtained from the combination of two mechanisms is always less than the product of the DA (or MA) of the two original mechanisms.

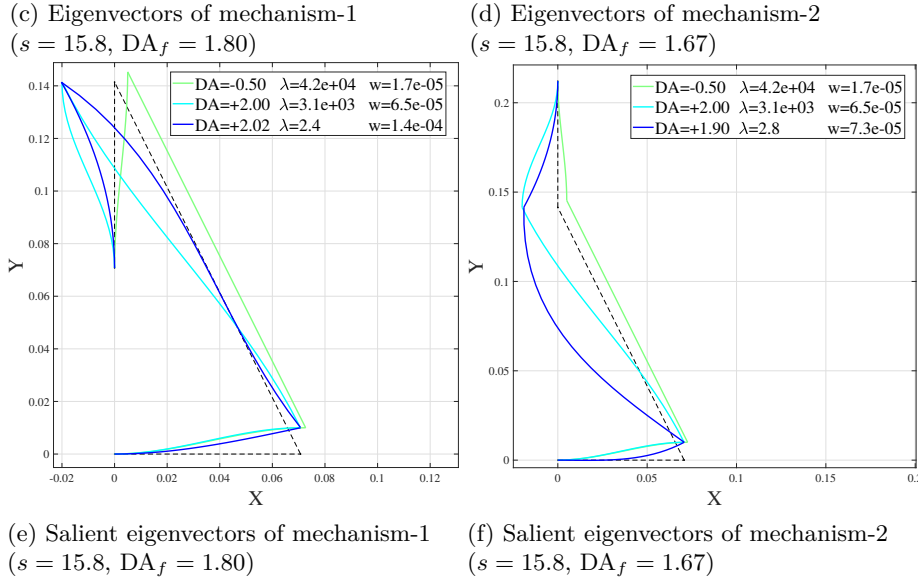
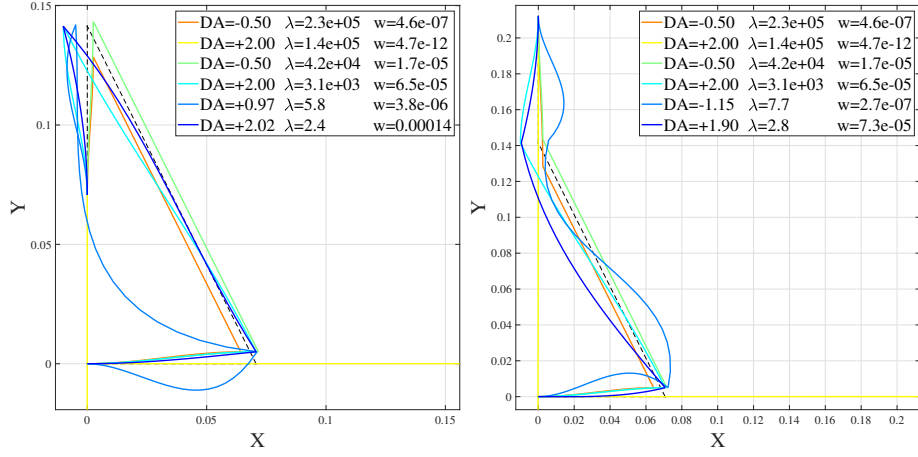
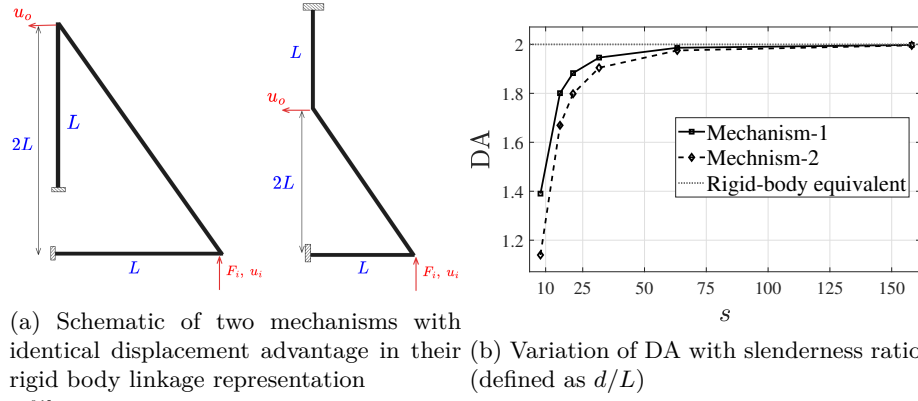


Fig. 2: Displacement advantage of mechanisms with identical (DA_{IC}) (Eigenvectors scaled to have uniform input displacement and w corresponds to the weight $\left(\frac{(a_k)_i^2}{\lambda_k}\right)$ of each mode shape)

4 Conclusions

In this work, the mechanical advantage of a compliant mechanism is expressed in terms of the kinetoelastics of the mechanism. The mechanical advantage of a compliant mechanism subject to large deformation is broken down into the mechanical advantage of a structure and the loss in mechanical advantage caused by the displacement of the output port necessary to engage the workpiece. The relations derived for a general deformation are simplified for infinitesimal deformation, and the results are then posed in terms of the eigenvectors and eigenvalues of the initial stiffness matrix. The interpretation of mechanical advantage through eigenvalues and eigenvectors is impractical for systems with a large degree of freedom, but can help interpret the characteristics for systems with few degrees of freedom, such as dyads, compliant-4-bar linkages, which are commonly employed in building block-based synthesis methods. Finally, it is shown that for small deformations, MA and DA values of a composition of compliant mechanisms are always less than the product of the respective values of the mechanisms that make up the composition.

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